

Module 2 Superconductivity Extension: Electrostatics in the Superconducting Regime

Textbook-Style Physics Note for RadiationEffects_proj2

How the Module 2 Poisson/electric-field model changes when superconducting films, condensate screening, quasiparticles, and radiation-induced charge perturbations are introduced

Purpose of this note. The original Module 2 solves electrostatic Poisson physics in a semiconductor-like silicon device: radiation-induced defect charge, dopant charge, and mobile electron/hole charge determine the potential $\phi(x, y)$ and electric field $\mathbf{E} = -\nabla\phi$. This note derives how that picture changes when the device is operated in the superconducting regime. The main correction is not that ordinary silicon automatically becomes superconducting. The realistic superconducting-device geometry is a hybrid structure: a silicon or dielectric substrate, oxide/interface layers, and superconducting metal films such as Al, Nb, Ta, or related multilayers. In this regime, Module 2 remains an electrostatic module, but the charge response and boundary conditions change qualitatively. Superconducting regions strongly screen electric fields, behave approximately as equipotential islands in the static limit, support condensate and quasiparticle degrees of freedom, and convert radiation-induced energy deposition into offset-charge shifts, trapped charge, pair-breaking phonons, and quasiparticle poisoning pathways.

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Symbols and acronyms used in this document

Symbol / acronym	Meaning	Typical units
Ω	Full computational domain	m^2 or m^3
Ω_d	Dielectric, semiconductor, substrate, or oxide region	m^2 or m^3
Ω_{sc}	Superconducting film or superconducting island region	m^2 or m^3
Γ_{int}	Interface between dielectric/semiconductor and superconductor	m or m^2
ϕ	Electrostatic scalar potential	V
ϕ_{sc}	Constant potential of a superconducting island	V
$\mathbf{E}$	Electric field, $\mathbf{E} = -\nabla\phi$ in the electrostatic limit	V/m
$\mathbf{D}$	Electric displacement field, $\mathbf{D} = \epsilon\mathbf{E}$ in a linear dielectric	C/m ²
ρ	Total charge density entering Gauss's law	C/m ³

Symbol / acronym	Meaning	Typical units
ρ_{ext}	External or fixed charge: defects, trapped charge, dopants, ionization charge	C/m ³
ρ_{ind}	Induced screening charge from mobile metallic or superconducting degrees of freedom	C/m ³
ρ_{trap}	Trapped charge density in substrate, oxide, or interface states	C/m ³
ρ_{def}	Radiation-induced defect charge density	C/m ³
ρ_{qp}	Quasiparticle charge-imbalance density	C/m ³
ε	Permittivity	F/m
ε_{d}	Effective dielectric/substrate permittivity	F/m
ε_{sc}	Effective permittivity used inside a screened superconductor model	F/m
λ_{scr}	Electrostatic screening length in a superconducting/metallic region	m
λ_{TF}	Thomas-Fermi screening length	m
λ_L	London magnetic penetration depth	m
Δ	Superconducting energy gap	J or eV
T_c	Critical temperature	K
$\Psi = \Psi e^{i\theta}$	Ginzburg-Landau order parameter	material dependent
θ	Superconducting phase	rad
μ	Chemical potential	J
μ_{ec}	Electrochemical potential	J
n_s	Superfluid or condensate electron density, depending on convention	m ⁻³
n_{qp}	Quasiparticle density	m ⁻³
$\Phi_0 = h/(2e)$	Superconducting flux quantum	Wb
V_i	Electrostatic potential of superconducting island i	V
Q_i	Net charge on superconducting island i	C
$\mathbf{C}$	Capacitance matrix for island network	F
N_i	FEM shape function	—

Symbol / acronym	Meaning	Typical units
K_{ij}	FEM stiffness matrix contribution	problem dependent
M_{ij}	FEM mass-like screening matrix contribution	problem dependent

1 Where this note fits in the project

The present project began with a semiconductor radiation-effects chain:

$$\text{energy deposition} \rightarrow \text{defect evolution} \rightarrow \text{electrostatics} \rightarrow \text{carrier transport} \rightarrow \text{device response.} \quad (1.1)$$

Module 2 occupies the electrostatic step. In its normal-state form, it solves

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho, \quad \mathbf{E} = -\nabla \phi, \quad (1.2)$$

with a semiconductor charge density of the form

$$\rho = e(p - n + N_D^+ - N_A^-) + \rho_{\text{def}}. \quad (1.3)$$

This is a natural model when the active material is silicon treated as a semiconductor with electrons, holes, dopants, and charged defect states.

A superconducting-device extension changes the physical interpretation of the electrostatic problem. At cryogenic temperature, a superconducting film is not just another semiconductor region with ordinary n and p . It is a charged quantum condensate plus a quasiparticle population, sitting on or near a dielectric or semiconducting substrate. Its static electric response is closer to a metal than to a depleted semiconductor: it screens electric fields over an atomic-to-nanometric length scale, and most of its bulk acts as an equipotential conductor. The substrate, oxide, and interface still hold trapped charge and radiation-induced charge. Those charges alter the potential landscape seen by the superconducting island, shift offset charge, and can modulate device response.

Main conceptual shift. In the normal Module 2 model, charge density in the active silicon volume creates the field. In the superconducting extension, the superconductor usually does *not* allow a large static electric field inside its bulk. Instead, external charge in the substrate/oxide is screened by surface and near-surface charge in the superconducting film. Therefore, Module 2 becomes partly a Poisson problem in the dielectric/substrate and partly a conductor/screening/capacitance problem for superconducting islands.

1.1 The realistic geometry: hybrid superconducting device

A typical superconducting quantum-device geometry is not pure superconducting silicon. It is closer to

$$\text{vacuum/air} \mid \text{superconducting metal film} \mid \text{native oxide/interface} \mid \text{silicon or sapphire substrate.} \quad (1.4)$$

Radiation can deposit energy in any of these regions. Module 1 can produce spatial energy-deposition maps. Module 3 can represent radiation-created defects, traps, or charge centers. Module 4 will later need phonon and thermal transport in the cryogenic regime. Module 5 will later need a superconducting replacement for ordinary drift-diffusion carrier transport. But Module 2 is the first place where the low-temperature superconducting environment changes the mathematics: the potential is no longer determined by ordinary semiconductor charge alone.

vacuum / package environment
superconducting film or island: screened, nearly equipotential
oxide/interface: traps, TLS-like charge fluctuators, fixed surface charge
substrate: dielectric or semiconductor Poisson region with radiation-induced charge

$$\text{Module 1 energy deposition and Module 3 defect/trap source} \longrightarrow \text{Module 2 } \{\phi, \mathbf{E}, Q_{\text{off}}\}. \quad (1.5)$$

2 Recap: ordinary Module 2 Poisson physics

The original Module 2 note begins from Gauss's law in matter,

$$\nabla \cdot \mathbf{D} = \rho, \quad \mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{E} = -\nabla \phi. \quad (2.1)$$

Substitution gives

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho. \quad (2.2)$$

The source term is built from mobile carriers, ionized dopants, and charged defect populations:

$$\rho = e(p - n + N_D^+ - N_A^-) + e \sum_a z_a C_a. \quad (2.3)$$

Here C_a is the concentration of radiation-induced defect species a and z_a is its effective charge number. In the normal semiconductor model, the unknown potential affects carrier density and transport. The electric field obtained from Module 2 is then passed to Module 5:

$$\phi \longrightarrow \mathbf{E} = -\nabla \phi \longrightarrow \text{drift-diffusion currents.} \quad (2.4)$$

This framework remains valid in non-superconducting regions of the superconducting-device geometry. The substrate, the oxide, and the vacuum gap still obey Gauss's law. What changes is the constitutive response inside the superconducting region and the boundary condition imposed by the superconducting metal.

3 What actually changes below the superconducting transition

3.1 Superconductivity does not replace Maxwell's equations

The first correction is conceptual. Superconductivity does not invalidate Gauss's law. The electrostatic equation remains

$$\nabla \cdot \mathbf{D} = \rho. \quad (3.1)$$

What changes is the relation between ρ and ϕ in the superconducting material. In a semiconductor, ρ is often expressed through carrier densities $n(\phi)$ and $p(\phi)$ plus dopants and defects. In a superconductor, most electronic charge belongs to a condensate that rearranges to keep the interior nearly charge neutral and the electrochemical potential nearly uniform. A static electric field is therefore screened out of the bulk.

Thus the correct question for Module 2 is not

$$\text{“Does Poisson’s equation still hold?”} \quad (3.2)$$

but rather

$$\text{“What charge response should be inserted into Poisson’s equation in a superconducting region?”} \quad (3.3)$$

3.2 Three regions must be separated

A practical superconducting-device Module 2 should distinguish at least three classes of regions:

1. **Dielectric or substrate regions** Ω_d . These still use Poisson's equation with fixed/trapped/defect charge sources.
2. **Superconducting metal regions** Ω_{sc} . These are either approximated as equipotential conductors or resolved with a screened-Poisson equation.
3. **Interfaces and tunnel barriers**. These are thin regions where trapped charge, oxide defects, electric fields, and Josephson-junction capacitance matter strongly.

This separation is not cosmetic. It determines whether a charge source creates a field throughout a region, is screened locally, or appears only as an effective offset charge on a superconducting island.

4 Electrochemical potential and why superconductors screen electric fields

4.1 Electrostatic potential versus electrochemical potential

For charged particles, the relevant equilibrium quantity is not only the chemical potential μ . It is the electrochemical potential. For electrons with charge $-e$, a common convention is

$$\mu_{\text{ec}} = \mu - e\phi. \quad (4.1)$$

At equilibrium, μ_{ec} is spatially constant:

$$\nabla\mu_{\text{ec}} = 0. \quad (4.2)$$

Therefore,

$$\nabla\mu = e\nabla\phi = -e\mathbf{E}. \quad (4.3)$$

A spatially varying electrostatic potential must be balanced by a spatially varying chemical potential, which means a spatial redistribution of electronic density.

For a metal or superconductor, the electronic compressibility is large. A small potential perturbation causes a redistribution of charge that screens the perturbation. This is the origin of short-range electrostatic screening.

4.2 Linear screening derivation

Let n denote the electronic density responding to a weak electrostatic perturbation. Linearize the chemical potential around equilibrium:

$$\delta\mu = \left(\frac{\partial\mu}{\partial n}\right)_0 \delta n. \quad (4.4)$$

Equilibrium requires $\delta\mu_{\text{ec}} = 0$, so from Eq. (4.1),

$$\delta\mu - e\delta\phi = 0. \quad (4.5)$$

Thus

$$\delta n = \left(\frac{\partial n}{\partial\mu}\right)_0 e\delta\phi. \quad (4.6)$$

The induced charge density from electrons is

$$\rho_{\text{ind}} = -e\delta n = -e^2 \left(\frac{\partial n}{\partial\mu}\right)_0 \delta\phi. \quad (4.7)$$

Define the screening length by

$$\frac{\varepsilon}{\lambda_{\text{scr}}^2} = e^2 \left(\frac{\partial n}{\partial \mu} \right)_0. \quad (4.8)$$

Then

$$\rho_{\text{ind}} = -\frac{\varepsilon}{\lambda_{\text{scr}}^2} \phi, \quad (4.9)$$

where the equilibrium reference potential has been chosen as $\phi = 0$.

Substitute $\rho = \rho_{\text{ext}} + \rho_{\text{ind}}$ into Poisson's equation:

$$\nabla \cdot (\varepsilon \nabla \phi) = -\rho_{\text{ext}} - \rho_{\text{ind}} = -\rho_{\text{ext}} + \frac{\varepsilon}{\lambda_{\text{scr}}^2} \phi. \quad (4.10)$$

Rearranging gives the screened-Poisson equation

$$\boxed{\nabla \cdot (\varepsilon \nabla \phi) - \frac{\varepsilon}{\lambda_{\text{scr}}^2} \phi = -\rho_{\text{ext}}.} \quad (4.11)$$

For uniform ε ,

$$\boxed{\nabla^2 \phi - \frac{1}{\lambda_{\text{scr}}^2} \phi = -\frac{\rho_{\text{ext}}}{\varepsilon}.} \quad (4.12)$$

This is a Helmholtz-type electrostatic equation. Unlike ordinary Poisson's equation, a local charge perturbation does not create a long-range potential inside the superconductor. It is screened over λ_{scr} .

Important distinction: electrostatic versus magnetic screening. The electrostatic screening length λ_{scr} is not the London penetration depth λ_L . The London penetration depth controls how magnetic field penetrates a superconductor. Electrostatic potential in a metal or superconductor is screened on a Thomas-Fermi-like length scale, usually much shorter than λ_L . For Module 2, which is electrostatic, the relevant screening is electric/charge screening, not magnetic Meissner screening.

4.3 Equipotential limit

If λ_{scr} is much smaller than the mesh scale or film thickness, then resolving Eq. (4.11) is numerically inefficient and physically unnecessary for a device-scale electrostatic solver. In the limit

$$\lambda_{\text{scr}} \ll h_{\text{mesh}}, \quad \lambda_{\text{scr}} \ll t_{\text{film}}, \quad (4.13)$$

the superconductor can be treated as an equipotential conductor:

$$\phi(\mathbf{x}) = V_j, \quad \mathbf{x} \in \Omega_{\text{sc},j}, \quad (4.14)$$

where V_j is the island potential of superconducting region j . The electric field is then zero in the superconducting bulk,

$$\mathbf{E} \approx \mathbf{0}, \quad \mathbf{x} \in \Omega_{\text{sc},j} \text{ away from surfaces and junction barriers.} \quad (4.15)$$

The required screening charge appears as surface charge at the boundary.

5 Condensate, quasiparticles, and charge density

5.1 BCS picture in the minimal language needed for Module 2

Below T_c , the electronic ground state is not a normal Fermi sea. Electrons near the Fermi surface form correlated Cooper pairs. The low-energy excitation spectrum has an energy gap Δ . In the simplest BCS weak-coupling limit, the zero-temperature gap obeys approximately

$$\Delta(0) \approx 1.76 k_B T_c. \quad (5.1)$$

The minimum energy to break a Cooper pair and create two quasiparticles is approximately

$$E_{\text{pair break}} \approx 2\Delta. \quad (5.2)$$

This matters for radiation effects because energy deposition can generate high-energy phonons or photons with energy above 2Δ , which can break Cooper pairs and create quasiparticles.

For Module 2, the BCS picture changes the charge model in two ways:

1. The condensate is a stiff charged quantum fluid that screens electric fields.
2. Quasiparticles are the remaining excitations above the superconducting ground state. Their equilibrium density is exponentially small at $T \ll T_c$, but radiation can produce nonequilibrium quasiparticles.

5.2 Equilibrium quasiparticle density

At low temperature, the equilibrium quasiparticle density has the qualitative form

$$n_{\text{qp}}^{\text{eq}}(T) \propto N_0 \sqrt{\Delta k_B T} \exp\left(-\frac{\Delta}{k_B T}\right), \quad (5.3)$$

where N_0 is the normal-state electronic density of states near the Fermi level. The exact prefactor depends on convention, spin degeneracy, and material model. The essential physics is the exponential factor. Deep below T_c , thermal quasiparticles should be rare. Therefore, a large quasiparticle population in a superconducting device is usually a nonequilibrium population.

5.3 Quasiparticle charge is not the same as ordinary semiconductor charge

In a semiconductor, electrons and holes carry definite charges and drift/diffuse in a band structure. In a superconductor, quasiparticles are Bogoliubov excitations: coherent mixtures of electron-like and hole-like components. A quasiparticle at energy E has coherence factors u and v , and its effective charge expectation depends on electron-hole imbalance. Near the gap edge, quasiparticle charge can be small or sign-dependent in a way that is not captured by ordinary electron/hole drift-diffusion.

For Module 2, this means we should not blindly replace Eq. (1.3) by

$$\rho = -en_{\text{qp}}. \quad (5.4)$$

That would be physically wrong in general. A safer reduced electrostatic representation is

$$\rho_{\text{qp}} = q_{\text{qp}}^{\text{eff}} n_{\text{qp}}^{\text{imb}}, \quad (5.5)$$

where $n_{\text{qp}}^{\text{imb}}$ is a charge-imbalance component of the quasiparticle distribution and $q_{\text{qp}}^{\text{eff}}$ is an effective charge parameter. If the quasiparticle population is electron-hole symmetric, it may strongly affect dissipation and qubit relaxation while contributing little to static space charge.

Practical consequence. In a first superconducting Module 2 extension, quasiparticles should usually enter indirectly through device-response parameters, island parity, or coupling to Module 5, not as a large ordinary Poisson charge density. Static electrostatic offsets are more likely to come from trapped substrate/oxide charge and surface charge than from a neutral quasiparticle population.

6 Ginzburg-Landau viewpoint and electromagnetic variables

6.1 Order parameter and free energy

A useful macroscopic description of a superconductor is the Ginzburg-Landau order parameter

$$\Psi(\mathbf{x}) = |\Psi(\mathbf{x})| e^{i\theta(\mathbf{x})}. \quad (6.1)$$

The simplest Ginzburg-Landau free energy density is

$$f = \alpha |\Psi|^2 + \frac{\beta}{2} |\Psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla - q^* \mathbf{A}) \Psi|^2 + \frac{|\mathbf{B}|^2}{2\mu_0}, \quad (6.2)$$

where $q^* = -2e$ and m^* are the Cooper-pair charge and effective mass. Varying this functional with respect to Ψ^* gives the first Ginzburg-Landau equation,

$$\alpha\Psi + \beta|\Psi|^2\Psi + \frac{1}{2m^*}(-i\hbar\nabla - q^*\mathbf{A})^2\Psi = 0. \quad (6.3)$$

Varying with respect to $\mathbf{A}$ gives the superconducting current density,

$$\mathbf{J}_s = \frac{q^*\hbar}{2m^*i}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) - \frac{(q^*)^2}{m^*}|\Psi|^2\mathbf{A}. \quad (6.4)$$

If $|\Psi|$ is approximately constant, this can be written as

$$\mathbf{J}_s = \frac{q^*|\Psi|^2}{m^*}(\hbar\nabla\theta - q^*\mathbf{A}). \quad (6.5)$$

This equation is mostly magnetic/phase physics, but it is important context for Module 2 because the electrostatic potential is connected to time evolution of the superconducting phase.

6.2 Gauge-invariant phase relation and voltage

The Josephson voltage relation connects voltage to the time derivative of superconducting phase difference:

$$V = \frac{\hbar}{2e} \frac{d\varphi}{dt}, \quad (6.6)$$

where φ is the gauge-invariant phase difference across a junction. Therefore, a truly static electrostatic solution with no DC voltage across a superconducting element corresponds to a time-independent phase difference:

$$V = 0 \quad \Rightarrow \quad \frac{d\varphi}{dt} = 0. \quad (6.7)$$

A DC voltage across a Josephson junction is not a static electrostatic state of the superconducting phase; it drives phase evolution. This is why a static Module 2 Poisson model can compute capacitance fields and offset potentials, but it should not be treated as a full Josephson dynamics solver.

7 Reformulating Module 2 for superconducting geometries

7.1 Domain decomposition

Split the computational domain into dielectric/substrate regions and superconducting regions:

$$\Omega = \Omega_d \cup \Omega_{sc}, \quad \Omega_d \cap \Omega_{sc} = \emptyset. \quad (7.1)$$

The dielectric region may include silicon substrate, oxide, vacuum gaps, and tunnel barriers. The superconducting region may include films, islands, resonators, ground planes, or leads.

In Ω_d , the electrostatic equation remains an ordinary Poisson equation:

$$\boxed{\nabla \cdot (\epsilon_d \nabla \phi_d) = -\rho_d, \quad \mathbf{x} \in \Omega_d.} \quad (7.2)$$

The source may include

$$\rho_d = \rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{dop}} + \rho_{\text{ion}} + \rho_{\text{mobile}}, \quad (7.3)$$

where the last terms are included only if the substrate is being treated as a semiconductor rather than a pure dielectric.

In Ω_{sc} , two modeling options are useful.

7.2 Model A: equipotential superconducting island

The simplest superconducting approximation is

$$\boxed{\phi_s(\mathbf{x}) = V_j, \quad \mathbf{x} \in \Omega_{\text{sc},j}.} \quad (7.4)$$

The potential V_j is constant over island j . If the island is connected to an ideal voltage source, V_j is prescribed. If it is floating, V_j is an unknown determined by charge neutrality or capacitance constraints.

This is usually the right first model for Module 2 because device-scale meshes cannot resolve the microscopic electric screening length. It also matches common electrostatic capacitance modeling of superconducting circuits.

7.3 Model B: screened-Poisson superconductor

If one wants to resolve charge penetration near a superconducting surface, use

$$\boxed{\nabla \cdot (\epsilon_{\text{sc}} \nabla \phi_s) - \frac{\epsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi_s - V_j) = -\rho_{s,\text{ext}}, \quad \mathbf{x} \in \Omega_{\text{sc},j}.} \quad (7.5)$$

Here V_j is the reference island potential and $\rho_{s,\text{ext}}$ denotes any externally imposed charge inside the superconducting material. In most device-scale radiation problems, $\rho_{s,\text{ext}}$ should be small or treated as localized near interfaces; the major external charges are in the substrate, oxide, or barrier.

Equation (7.5) forces ϕ_s back toward V_j over the screening length λ_{scr} . In the limit $\lambda_{\text{scr}} \rightarrow 0$, it reduces to Model A.

8 Interface conditions

8.1 Potential continuity

At an ideal interface without an explicit dipole sheet, electrostatic potential is continuous:

$$\boxed{\phi_d = \phi_s, \quad \mathbf{x} \in \Gamma_{\text{int}}.} \quad (8.1)$$

If an interface dipole layer or work-function offset is represented explicitly, the model may instead use a potential jump:

$$\phi_s - \phi_d = \Delta\phi_{\text{dipole}}. \quad (8.2)$$

For the first project implementation, potential continuity is cleaner.

8.2 Displacement-field jump

The normal component of $\mathbf{D}$ jumps by the free surface charge density:

$$\boxed{\mathbf{n} \cdot (\mathbf{D}_d - \mathbf{D}_s) = \sigma_{\text{int}}, \quad \mathbf{x} \in \Gamma_{\text{int}},} \quad (8.3)$$

where $\mathbf{n}$ is a chosen normal and σ_{int} is interface charge density. With $\mathbf{D} = -\varepsilon \nabla \phi$,

$$\mathbf{n} \cdot (-\varepsilon_d \nabla \phi_d + \varepsilon_{sc} \nabla \phi_s) = \sigma_{\text{int}}. \quad (8.4)$$

In the equipotential-conductor limit, $\nabla \phi_s \approx 0$ inside the superconductor, and the interface charge becomes the surface charge required to terminate the dielectric field:

$$\sigma_{sc} \approx -\mathbf{n} \cdot \mathbf{D}_d. \quad (8.5)$$

8.3 Thin-film and surface-charge interpretation

A superconducting film can be much thinner than the lateral dimensions of the device. If the film thickness is not resolved, it can be represented as a conducting boundary with surface charge. The dielectric/substrate Poisson problem is then solved with Dirichlet boundary conditions on superconducting islands:

$$\phi_d = V_j \quad \text{on the surface of island } j. \quad (8.6)$$

The induced charge on the island is computed afterward from the electric displacement field:

$$Q_j = \int_{\Gamma_j} \sigma_j dA = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon_d \nabla \phi_d dA. \quad (8.7)$$

This is often more useful for superconducting circuits than the bulk potential itself, because island charge determines offset charge and capacitance energy.

9 Floating superconducting islands and capacitance constraints

9.1 Fixed-potential versus floating-potential islands

A superconducting region connected to a voltage source has prescribed potential:

$$V_j = V_j^{\text{bias}}. \quad (9.1)$$

A floating island is different. Its potential is not externally fixed. It is determined by total charge balance. If island j has specified net charge Q_j^{given} , then

$$Q_j^{\text{given}} = - \int_{\Gamma_j} \mathbf{n} \cdot \epsilon \nabla \phi dA. \quad (9.2)$$

This integral constraint replaces a simple Dirichlet condition. The unknown V_j becomes an additional degree of freedom.

9.2 Capacitance matrix form

For multiple superconducting islands, electrostatics can often be reduced to a capacitance matrix relation:

$$Q_i = \sum_j C_{ij} V_j + Q_{i,\text{off}}. \quad (9.3)$$

Here $Q_{i,\text{off}}$ is offset charge induced by trapped substrate/oxide charge, radiation-generated charge, or nearby defects. More explicitly, one can compute capacitance by solving basis electrostatic problems:

$$\phi^{(j)} = 1 \text{ on island } j, \quad \phi^{(j)} = 0 \text{ on all other conductors}, \quad (9.4)$$

and then integrating displacement flux to obtain C_{ij} .

The electrostatic energy is

$$U = \frac{1}{2} \int_{\Omega_d} \epsilon |\nabla \phi|^2 dV = \frac{1}{2} \mathbf{V}^T \mathbf{C} \mathbf{V} \quad (9.5)$$

for the ideal conductor problem without fixed space charge. With offset charges, circuit models often rewrite the energy in charge variables. For a single island with total capacitance C_Σ ,

$$U = \frac{(Q - Q_{\text{off}})^2}{2C_\Sigma}. \quad (9.6)$$

In superconducting qubit notation, $Q = 2eN$ and $Q_{\text{off}} = 2en_g$, so the charging energy becomes

$$U = 4E_C(N - n_g)^2, \quad E_C = \frac{e^2}{2C_\Sigma}. \quad (9.7)$$

This is the direct bridge between Module 2 electrostatics and superconducting-device response: radiation-induced trapped charge changes Q_{off} , which changes n_g .

Module 2 output changes. In the normal semiconductor version, Module 2 primarily outputs $\phi(x, y)$ and $\mathbf{E}(x, y)$. In a superconducting-device version, Module 2 should additionally output island charge, capacitance, and offset charge:

$$\phi, \mathbf{E}, Q_j, C_{ij}, Q_{j,\text{off}}, n_{g,j} = Q_{j,\text{off}}/(2e). \quad (9.8)$$

10 Weak form for the superconducting Module 2 equations

10.1 Ordinary dielectric Poisson weak form

In a dielectric/substrate region Ω_d , multiply Eq. (7.2) by a test function v and integrate:

$$\int_{\Omega_d} v \nabla \cdot (\varepsilon_d \nabla \phi) d\Omega = - \int_{\Omega_d} v \rho_d d\Omega. \quad (10.1)$$

Integrating by parts gives

$$- \int_{\Omega_d} \varepsilon_d \nabla v \cdot \nabla \phi d\Omega + \int_{\partial\Omega_d} v \varepsilon_d \nabla \phi \cdot \mathbf{n} d\Gamma = - \int_{\Omega_d} v \rho_d d\Omega. \quad (10.2)$$

Multiplying by -1 gives the standard weak form:

$$\boxed{\int_{\Omega_d} \varepsilon_d \nabla v \cdot \nabla \phi d\Omega = \int_{\Omega_d} v \rho_d d\Omega + \int_{\partial\Omega_d} v (-\varepsilon_d \nabla \phi \cdot \mathbf{n}) d\Gamma.} \quad (10.3)$$

Dirichlet conditions are applied strongly or by penalty/Nitsche methods. Neumann fluxes enter through the boundary integral.

10.2 Screened-Poisson weak form

For a screened superconductor, start from

$$\nabla \cdot (\varepsilon_{\text{sc}} \nabla \phi_s) - \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi_s - V_j) = -\rho_{s,\text{ext}}. \quad (10.4)$$

Multiply by v and integrate over $\Omega_{\text{sc},j}$:

$$\int v \nabla \cdot (\varepsilon_{\text{sc}} \nabla \phi_s) d\Omega - \int v \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi_s - V_j) d\Omega = - \int v \rho_{s,\text{ext}} d\Omega. \quad (10.5)$$

After integration by parts and multiplying by -1 ,

$$\boxed{\int_{\Omega_{\text{sc},j}} \varepsilon_{\text{sc}} \nabla v \cdot \nabla \phi_s d\Omega + \int_{\Omega_{\text{sc},j}} v \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi_s - V_j) d\Omega = \int_{\Omega_{\text{sc},j}} v \rho_{s,\text{ext}} d\Omega + \text{boundary terms.}} \quad (10.6)$$

The new term is a mass-like screening term. With finite-element basis functions N_i and N_j , the element matrix becomes

$$K_{ij}^{(e)} = \int_{\Omega_e} \varepsilon \nabla N_i \cdot \nabla N_j d\Omega + \int_{\Omega_e} \frac{\varepsilon}{\lambda_{\text{scr}}^2} N_i N_j d\Omega. \quad (10.7)$$

The right-hand side gains

$$b_i^{(e)} = \int_{\Omega_e} N_i \rho_{\text{ext}} d\Omega + \int_{\Omega_e} \frac{\varepsilon}{\lambda_{\text{scr}}^2} N_i V_j d\Omega. \quad (10.8)$$

This is a direct code-level change relative to the existing Module 2 Poisson FEM implementation.

10.3 Equipotential island as a constrained FEM problem

If a superconducting island is treated as an equipotential conductor, all nodes on the island share a single unknown V_j rather than independent nodal potentials. There are three practical ways to implement this:

1. **Prescribed island potential.** Apply Dirichlet condition $\phi = V_j$ on the island boundary.
2. **Floating island with charge constraint.** Add unknown V_j and enforce Eq. (9.2) as a global constraint.
3. **Capacitance postprocessing.** Solve multiple basis problems with prescribed island voltages, compute the capacitance matrix, then evaluate floating potentials and offset charge in circuit form.

The third approach is usually the cleanest first step because it separates field solving from circuit interpretation.

11 How radiation-induced charge enters superconducting electrostatics

11.1 Trapped charge and offset charge

Radiation creates electron-hole pairs, phonons, excitations, and defect states. At cryogenic temperature, many carriers may become trapped rather than thermally released. Trapped charge in a substrate or oxide enters Module 2 as

$$\rho_{\text{trap}}(\mathbf{x}, t) = e \sum_a z_a C_a^{\text{trap}}(\mathbf{x}, t). \quad (11.1)$$

This charge may be spatially separated from a superconducting island but capacitively coupled to it. The induced offset charge on island j can be written schematically as a linear functional of trapped charge:

$$Q_{j,\text{off}}(t) = \int_{\Omega_d} W_j(\mathbf{x}) \rho_{\text{trap}}(\mathbf{x}, t) d\Omega, \quad (11.2)$$

where $W_j(\mathbf{x})$ is a weighting potential determined by electrostatic geometry. A practical way to compute W_j is to solve the basis problem with island j at unit potential and all other conductors grounded. This is the same idea used in capacitance extraction.

11.2 Radiation-generated quasiparticles

Ionizing radiation or cosmic-ray-like events can generate energetic phonons in the substrate. If those phonons reach superconducting films with energy exceeding 2Δ , they can break Cooper pairs and produce quasiparticles. This pathway is not primarily an electrostatic Poisson source; it is a superconducting excitation and dissipation source. However, it affects Module 2 indirectly through two mechanisms:

1. Nonequilibrium quasiparticles can produce parity changes and effective offset-charge jumps on superconducting islands.
2. The same radiation event that generates quasiparticles can also create substrate/oxide charge that appears directly in ρ_{trap} .

Thus the electrostatic and quasiparticle problems are coupled but not identical.

11.3 Charge noise and interface defects

At oxide and substrate interfaces, defects may switch between charge configurations. A two-level fluctuator with charge displacement Δq creates a change in electrostatic potential. If the fluctuator couples to island j , it changes $n_{g,j}$. In reduced form,

$$\delta n_{g,j} = \frac{1}{2e} \int_{\Omega_d} W_j(\mathbf{x}) \delta \rho_{\text{TLS}}(\mathbf{x}) d\Omega. \quad (11.3)$$

This formula shows how Module 2 can become the bridge from microscopic radiation-induced/interface defects to superconducting qubit offset-charge noise.

12 Comparison with the normal Module 2 source model

The normal-state Module 2 charge density was

$$\rho_{\text{normal}} = e(p - n + N_D^+ - N_A^-) + \rho_{\text{def}}. \quad (12.1)$$

A superconducting-device charge model should instead be region dependent:

$$\rho(\mathbf{x}) = \begin{cases} \rho_d(\mathbf{x}) = \rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{dop}} + \rho_{\text{mobile}}, & \mathbf{x} \in \Omega_d, \\ \rho_s(\mathbf{x}) = \rho_{s,\text{ext}} - \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2}(\phi_s - V_j), & \mathbf{x} \in \Omega_{\text{sc},j}. \end{cases} \quad (12.2)$$

This equation captures the essential change. In the dielectric/substrate, external charge remains a source. In the superconductor, induced charge is primarily a response that screens the potential.

Physics item	Normal semiconductor Module 2	Superconducting Module 2 extension
Active charge carriers	Electrons and holes with densities n, p	Condensate plus quasiparticles; static bulk charge strongly screened
Main PDE in active material	Poisson equation with semiconductor charge density	Equipotential conductor or screened-Poisson equation
Defect charge role	Direct volume source in silicon	Direct source mainly in substrate/oxide/interface; screened inside superconducting metal
Boundary behavior	Contacts impose voltage or flux	Superconducting islands are equipotential; floating islands require charge constraints
Important outputs	$\phi(x, y)$ and $\mathbf{E}(x, y)$	ϕ , $\mathbf{E}$, island charge, capacitance, offset charge n_g
Radiation signature	Field distortion and recombination/trapping effects	Offset-charge jumps, trapped-charge fields, quasiparticle generation, parity events

13 Boundary conditions for Josephson junctions and tunnel barriers

13.1 Tunnel barrier as a dielectric capacitor

A Josephson junction contains two superconducting electrodes separated by a thin insulating barrier. Electrostatics across the barrier is ordinary dielectric electrostatics:

$$\nabla \cdot (\varepsilon_b \nabla \phi_b) = -\rho_b, \quad \mathbf{x} \in \Omega_b. \quad (13.1)$$

If the barrier is thin and approximately uniform, the junction capacitance is approximately

$$C_J \approx \frac{\varepsilon_b A_J}{d_b}, \quad (13.2)$$

where A_J is junction area and d_b is barrier thickness. The voltage across the junction is

$$V_J = V_1 - V_2. \quad (13.3)$$

The electrostatic energy stored in the junction capacitance is

$$U_C = \frac{1}{2} C_J V_J^2. \quad (13.4)$$

This is still Module 2 physics. Josephson tunneling and phase dynamics are not solved by static Poisson, but the capacitance and electric field energy are.

13.2 Static Poisson cannot replace Josephson dynamics

The Josephson current relation is

$$I_s = I_c \sin \varphi. \quad (13.5)$$

The voltage relation is Eq. (6.6). These equations involve the superconducting phase difference. Static Module 2 can supply capacitance, electric field energy, island voltages, and offset charge. It cannot by itself predict coherent Josephson oscillations, qubit transition frequencies, or phase-slip dynamics. Those belong to a later superconducting device-response model.

14 Nondimensionalization and numerical scale separation

14.1 Standard Poisson scaling

For a characteristic length L , potential scale ϕ_0 , and charge scale ρ_0 , define

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{L}, \quad \hat{\phi} = \frac{\phi}{\phi_0}, \quad \hat{\rho} = \frac{\rho}{\rho_0}. \quad (14.1)$$

Ordinary Poisson becomes

$$\hat{\nabla} \cdot (\hat{\varepsilon} \hat{\nabla} \hat{\phi}) = -\eta \hat{\rho}, \quad \eta = \frac{\rho_0 L^2}{\varepsilon_0 \phi_0}. \quad (14.2)$$

14.2 Screened-Poisson scaling

The screened-Poisson equation becomes

$$\hat{\nabla} \cdot (\hat{\varepsilon} \hat{\nabla} \hat{\phi}) - \hat{\varepsilon} \left(\frac{L}{\lambda_{\text{scr}}} \right)^2 \hat{\phi} = -\eta \hat{\rho}_{\text{ext}}. \quad (14.3)$$

The dimensionless number

$$S = \left(\frac{L}{\lambda_{\text{scr}}} \right)^2 \quad (14.4)$$

measures stiffness. For device-scale L and microscopic λ_{scr} , S can be enormous. This is why the equipotential-conductor approximation is not merely a convenience; it is the correct reduced model when the mesh cannot resolve the screening layer.

15 Recommended modeling hierarchy for this project

15.1 Level 0: keep original Module 2 for non-superconducting silicon

If the geometry is still purely silicon and no superconducting film or superconducting boundary is represented, keep the existing Module 2 equation:

$$\nabla \cdot (\varepsilon_{\text{Si}} \nabla \phi) = -\rho. \quad (15.1)$$

Do not claim superconductivity has been added merely because the temperature is low. Ordinary silicon at cryogenic temperature is not a conventional superconducting metal in the relevant device sense.

15.2 Level 1: superconducting islands as Dirichlet/equipotential conductors

Represent superconducting films as conductor boundaries:

$$\phi = V_j \quad \text{on superconducting island } j. \quad (15.2)$$

Solve Poisson only in the substrate, oxide, barrier, and vacuum regions. Compute induced island charge using Eq. (8.7). This is the most stable first extension.

15.3 Level 2: floating islands and offset charge

Add charge constraints for floating islands and compute offset charge from trapped radiation-induced charge:

$$Q_{j,\text{off}} = \int W_j(\mathbf{x}) \rho_{\text{trap}}(\mathbf{x}) d\Omega. \quad (15.3)$$

This connects radiation-induced defects to superconducting qubit variables.

15.4 Level 3: screened-Poisson superconducting regions

Use Eq. (7.5) only if the mesh and physics goals require resolving near-surface screening. In most finite-element device models, this level should be used cautiously because of severe scale separation.

15.5 Level 4: dynamic superconducting electrodynamics

If voltage pulses, microwave drive, Josephson phase dynamics, quasiparticle diffusion, or nonequilibrium charge imbalance are required, Module 2 alone is insufficient. The project then needs additional dynamic equations:

$$\text{Poisson/capacitance} + \text{Josephson phase dynamics} + \text{quasiparticle transport} + \text{phonon transport}. \quad (15.4)$$

This will belong partly to Module 4, Module 5, and Module 7 rather than only Module 2.

16 Implementation consequences for MATLAB Module 2

16.1 Data structures

A superconducting Module 2 implementation should add material/region labels:

$$\text{regionType} \in \{\text{dielectric}, \text{semiconductor}, \text{superconductor}, \text{barrier}\}. \quad (16.1)$$

Each region needs properties:

$$\epsilon_r, \quad \rho_{\text{fixed}}, \quad \lambda_{\text{scr}} \text{ if screened}, \quad V_j \text{ if conductor/equipotential}. \quad (16.2)$$

16.2 Matrix assembly changes

For ordinary Poisson elements, the local matrix is

$$K_{ij}^{(e)} = \int_{\Omega_e} \varepsilon \nabla N_i \cdot \nabla N_j d\Omega. \quad (16.3)$$

For screened superconducting elements, add

$$M_{ij}^{(e)} = \int_{\Omega_e} \frac{\varepsilon}{\lambda_{\text{scr}}^2} N_i N_j d\Omega. \quad (16.4)$$

Therefore,

$$K_{ij}^{(e)} \leftarrow K_{ij}^{(e)} + M_{ij}^{(e)}. \quad (16.5)$$

The right-hand side adds the reference-potential term

$$b_i^{(e)} \leftarrow b_i^{(e)} + \int_{\Omega_e} \frac{\varepsilon}{\lambda_{\text{scr}}^2} N_i V_j d\Omega. \quad (16.6)$$

For equipotential islands, no screening matrix is assembled inside the island; the island is imposed as a boundary or global constraint.

16.3 Postprocessing changes

The normal Module 2 postprocessing computes

$$\mathbf{E} = -\nabla\phi. \quad (16.7)$$

The superconducting extension should additionally compute

$$Q_j = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon \nabla \phi dA, \quad (16.8)$$

$$C_{ij} = \text{island charge response to basis voltage } V_j = 1, \quad (16.9)$$

$$n_{g,j} = \frac{Q_{j,\text{off}}}{2e}, \quad (16.10)$$

$$U_E = \frac{1}{2} \int \varepsilon |\nabla \phi|^2 dV. \quad (16.11)$$

These outputs are the bridge to superconducting circuit response.

17 What should not be overclaimed

Avoid these modeling mistakes.

1. Do not model a superconducting film as ordinary silicon with lower temperature.
2. Do not use the London penetration depth as the electrostatic screening length.
3. Do not insert total quasiparticle density as if every quasiparticle were an ordinary electron charge density in Poisson's equation.
4. Do not use static Poisson to model Josephson phase dynamics.
5. Do not resolve Thomas-Fermi screening with a device-scale mesh unless the project specifically targets microscopic surface physics.

18 Summary of the superconducting Module 2 equation set

The clean reduced model for Module 2 in a superconducting-device geometry is:

$$\nabla \cdot (\varepsilon_d \nabla \phi_d) = -\rho_d, \quad \mathbf{x} \in \Omega_d, \quad (18.1)$$

$$\phi_s = V_j, \quad \mathbf{x} \in \Omega_{sc,j} \quad (\text{equipotential model}), \quad (18.2)$$

$$\phi_d = \phi_s, \quad \mathbf{x} \in \Gamma_{int}, \quad (18.3)$$

$$\mathbf{n} \cdot (\mathbf{D}_d - \mathbf{D}_s) = \sigma_{int}, \quad \mathbf{x} \in \Gamma_{int}, \quad (18.4)$$

$$Q_j = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon_d \nabla \phi_d dA, \quad \text{island charge}, \quad (18.5)$$

$$n_{g,j} = \frac{Q_{j,off}}{2e}, \quad \text{dimensionless offset charge}. \quad (18.6)$$

If screening inside the superconductor is explicitly resolved, replace the equipotential equation with

$$\nabla \cdot (\varepsilon_{sc} \nabla \phi_s) - \frac{\varepsilon_{sc}}{\lambda_{scr}^2} (\phi_s - V_j) = -\rho_{s,ext}. \quad (18.7)$$

19 Connection to later superconducting notes

This note only modifies Module 2. It prepares the foundation for the next superconducting physics notes:

- **Module 3 superconductivity:** defect/trap evolution at cryogenic temperature, charge-state freeze-out, trap occupancy, interface fluctuators, and radiation-created defect populations

relevant to offset charge.

- **Module 4 superconductivity:** cryogenic phonon transport, pair-breaking phonons, quasiparticle generation, phonon downconversion, Kapitza/interface thermal resistance, and electron-phonon bottlenecks.
- **Module 5 superconductivity:** replacement of ordinary electron/hole drift-diffusion with quasiparticle transport, charge imbalance, recombination, trapping, parity switching, and coupling to superconducting circuit variables.

20 Minimal equations to carry into code

For implementation planning, the most useful compact set is:

$$\text{Dielectric/substrate:} \quad \nabla \cdot (\varepsilon \nabla \phi) = -(\rho_{\text{def}} + \rho_{\text{trap}} + \rho_{\text{mobile}}), \quad (20.1)$$

$$\text{Superconducting island:} \quad \phi = V_j \quad \text{or} \quad \nabla \cdot (\varepsilon \nabla \phi) - \frac{\varepsilon}{\lambda_{\text{scr}}^2}(\phi - V_j) = -\rho_{s,\text{ext}}, \quad (20.2)$$

$$\text{Island charge:} \quad Q_j = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon \nabla \phi \, dA, \quad (20.3)$$

$$\text{Offset charge:} \quad Q_{j,\text{off}} = \int W_j(\mathbf{x}) \rho_{\text{trap}}(\mathbf{x}) \, d\Omega, \quad (20.4)$$

$$\text{Qubit-relevant offset:} \quad n_{g,j} = Q_{j,\text{off}}/(2e). \quad (20.5)$$

Bottom line. The superconducting extension of Module 2 is not a replacement of Poisson's equation. It is a replacement of the charge-response model and boundary-condition model. Poisson remains the field equation in the dielectric/substrate. Superconductors enter as screened, nearly equipotential islands whose induced surface charge, capacitance, and offset-charge response are the important outputs.

21 Derivation of offset-charge weighting potentials

21.1 Why a trapped charge can be converted into an island charge

Suppose a trapped charge distribution $\rho_t(\mathbf{x})$ sits in the dielectric/substrate while superconducting islands are held at fixed potentials. The electrostatic potential satisfies

$$\nabla \cdot (\varepsilon \nabla \phi) = -\rho_t. \quad (21.1)$$

The induced charge on island j is a surface-flux integral,

$$Q_j = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon \nabla \phi \, dA. \quad (21.2)$$

Directly recomputing this for every possible trapped-charge distribution can be expensive. A more useful formulation uses a weighting potential.

Define $w_j(\mathbf{x})$ by the basis problem

$$\nabla \cdot (\varepsilon \nabla w_j) = 0, \quad \mathbf{x} \in \Omega_d, \quad (21.3)$$

$$w_j = 1, \quad \mathbf{x} \in \Gamma_j, \quad (21.4)$$

$$w_j = 0, \quad \mathbf{x} \in \Gamma_k, \quad k \neq j, \quad (21.5)$$

$$\mathbf{n} \cdot \varepsilon \nabla w_j = 0, \quad \text{on insulating external boundaries.} \quad (21.6)$$

This is a purely geometric electrostatic solve. Once w_j is known, Green's identity gives the induced island charge from trapped charge:

$$Q_{j,\text{off}} = \int_{\Omega_d} w_j(\mathbf{x}) \rho_t(\mathbf{x}) \, d\Omega, \quad (21.7)$$

up to the sign convention used for outward normals and island charge. The important point is that the same trapped charge has a different effect depending on where it is located relative to the island. The function w_j is the electrostatic sensitivity map.

21.2 Green's identity derivation

Let ϕ_t be the potential produced by trapped charge with all conductors grounded:

$$\nabla \cdot (\varepsilon \nabla \phi_t) = -\rho_t, \quad \phi_t = 0 \text{ on all conductor surfaces.} \quad (21.8)$$

Let w_j solve the weighting-potential problem. Green's second identity for the operator $\nabla \cdot \varepsilon \nabla$ gives

$$\int_{\Omega_d} [w_j \nabla \cdot (\varepsilon \nabla \phi_t) - \phi_t \nabla \cdot (\varepsilon \nabla w_j)] \, d\Omega = \int_{\partial\Omega_d} [w_j \varepsilon \nabla \phi_t \cdot \mathbf{n} - \phi_t \varepsilon \nabla w_j \cdot \mathbf{n}] \, dA. \quad (21.9)$$

Because $\nabla \cdot (\varepsilon \nabla w_j) = 0$ and $\nabla \cdot (\varepsilon \nabla \phi_t) = -\rho_t$, the left side is

$$- \int_{\Omega_d} w_j \rho_t \, d\Omega. \quad (21.10)$$

On conductor surfaces, $\phi_t = 0$, so the second boundary term vanishes. Since $w_j = 1$ on conductor j and zero on other grounded conductors, the boundary term reduces to

$$\int_{\Gamma_j} \varepsilon \nabla \phi_t \cdot \mathbf{n} dA. \quad (21.11)$$

With island charge defined as $Q_j = -\int_{\Gamma_j} \varepsilon \nabla \phi_t \cdot \mathbf{n} dA$, we obtain Eq. (21.7). This derivation is useful because it turns a field problem into a precomputed sensitivity function.

22 Variational interpretation of screened electrostatics

22.1 Energy functional

The screened-Poisson equation can also be derived by minimizing an electrostatic free-energy functional. For a superconducting region whose potential is screened toward a reference island potential V_j , define

$$\mathcal{F}[\phi] = \int_{\Omega_{\text{sc},j}} \left[\frac{1}{2} \varepsilon |\nabla \phi|^2 + \frac{1}{2} \frac{\varepsilon}{\lambda_{\text{scr}}^2} (\phi - V_j)^2 - \rho_{\text{ext}} \phi \right] d\Omega. \quad (22.1)$$

The first term is ordinary electric-field energy. The second term is the energetic penalty for charge imbalance away from the superconducting electrochemical equilibrium. The third term couples the potential to fixed external charge.

Take a variation $\phi \rightarrow \phi + \eta v$ and require

$$\left. \frac{d\mathcal{F}[\phi + \eta v]}{d\eta} \right|_{\eta=0} = 0. \quad (22.2)$$

The variation is

$$\delta\mathcal{F} = \int \left[\varepsilon \nabla \phi \cdot \nabla v + \frac{\varepsilon}{\lambda_{\text{scr}}^2} (\phi - V_j) v - \rho_{\text{ext}} v \right] d\Omega. \quad (22.3)$$

Integrating the first term by parts gives

$$\delta\mathcal{F} = \int \left[-\nabla \cdot (\varepsilon \nabla \phi) + \frac{\varepsilon}{\lambda_{\text{scr}}^2} (\phi - V_j) - \rho_{\text{ext}} \right] v d\Omega + \int_{\partial\Omega} v \varepsilon \nabla \phi \cdot \mathbf{n} dA. \quad (22.4)$$

For arbitrary v in the interior, the Euler-Lagrange equation is

$$-\nabla \cdot (\varepsilon \nabla \phi) + \frac{\varepsilon}{\lambda_{\text{scr}}^2} (\phi - V_j) - \rho_{\text{ext}} = 0, \quad (22.5)$$

which is equivalent to

$$\nabla \cdot (\varepsilon \nabla \phi) - \frac{\varepsilon}{\lambda_{\text{scr}}^2} (\phi - V_j) = -\rho_{\text{ext}}. \quad (22.6)$$

This confirms the weak form and the positive-definite FEM matrix contribution in Eq. (10.7).

22.2 Equipotential limit from the energy

The screening penalty term scales as

$$\int \frac{\varepsilon}{2\lambda_{\text{scr}}^2} (\phi - V_j)^2 d\Omega. \quad (22.7)$$

As $\lambda_{\text{scr}} \rightarrow 0$, any finite-energy solution must satisfy

$$\phi \rightarrow V_j \quad (22.8)$$

inside the superconducting region. Thus the equipotential conductor model is not an ad hoc shortcut. It is the asymptotic limit of the screened electrostatic free energy.

23 One-dimensional screening example

23.1 Potential decay into a superconducting half-space

Consider a flat superconducting half-space $x > 0$ with a surface potential perturbation $\phi(0) = \phi_0$ and no external volume charge. The screened-Poisson equation is

$$\frac{d^2\phi}{dx^2} - \frac{\phi}{\lambda_{\text{scr}}^2} = 0. \quad (23.1)$$

The solution that remains finite as $x \rightarrow \infty$ is

$$\phi(x) = \phi_0 e^{-x/\lambda_{\text{scr}}}. \quad (23.2)$$

The electric field is

$$E_x(x) = -\frac{d\phi}{dx} = \frac{\phi_0}{\lambda_{\text{scr}}} e^{-x/\lambda_{\text{scr}}}. \quad (23.3)$$

The induced charge density follows from Eq. (4.9):

$$\rho_{\text{ind}}(x) = -\frac{\varepsilon}{\lambda_{\text{scr}}^2} \phi_0 e^{-x/\lambda_{\text{scr}}}. \quad (23.4)$$

The integrated surface screening charge per unit area is

$$\sigma_{\text{scr}} = \int_0^\infty \rho_{\text{ind}}(x) dx = -\frac{\varepsilon\phi_0}{\lambda_{\text{scr}}}. \quad (23.5)$$

This simple solution explains why the bulk of a superconductor is field-free while the surface can carry the charge required to satisfy boundary conditions.

23.2 Numerical implication

If λ_{scr} is far smaller than the smallest mesh element, a finite-element solver cannot accurately represent $e^{-x/\lambda_{\text{scr}}}$. Attempting to do so creates a stiff, ill-conditioned problem. For a device-scale solver, replacing the screened layer by an equipotential conductor with surface charge is the correct reduced description.

24 How Module 2 superconductivity changes the PINN/CPINN interpretation

The existing Module 2 PINN and CPINN extensions use a neural approximation to the potential and penalize the Poisson residual. In the superconducting extension, the residual must become region dependent:

$$\mathcal{R}_P(\mathbf{x}) = \begin{cases} \nabla \cdot (\varepsilon_d \nabla \phi_\theta) + \rho_d, & \mathbf{x} \in \Omega_d, \\ \nabla \cdot (\varepsilon_{\text{sc}} \nabla \phi_\theta) - \frac{\varepsilon_{\text{sc}}}{\lambda_{\text{scr}}^2} (\phi_\theta - V_j) + \rho_{s,\text{ext}}, & \mathbf{x} \in \Omega_{\text{sc},j} \text{ if screened,} \\ \phi_\theta - V_j, & \mathbf{x} \in \Omega_{\text{sc},j} \text{ if equipotential penalty is used.} \end{cases} \quad (24.1)$$

Boundary/interface losses must also change. A superconducting island is not just another boundary segment with arbitrary Neumann flux. Its potential is constrained, and its net charge may be a global output or constraint.

For CPINN specifically, the causal or continuation coordinate should not be interpreted as physical time for a static Poisson problem. Better ordered coordinates are

$$\alpha \in [0, 1] \quad (24.2)$$

where α may represent increasing trapped-charge amplitude, increasing radiation dose, increasing defect population, or increasing coupling iteration index. The superconducting version can use a continuation from zero trapped charge to full radiation-induced trapped charge:

$$\rho_t(\mathbf{x}; \alpha) = \alpha \rho_t^{\text{final}}(\mathbf{x}). \quad (24.3)$$

The model then learns how island offset charge and field patterns evolve as radiation-induced charge is introduced.

25 Verification tests for a future superconducting Module 2 code

A future MATLAB implementation should not begin with the full hybrid radiation problem. It should pass simple tests first.

25.1 Test 1: grounded plane above a point-like trapped charge

Place a localized charge in the dielectric below a grounded superconducting plane. The plane should remain $\phi = 0$, and induced charge should appear on the plane. The field pattern should resemble an image-charge solution in the appropriate geometry.

25.2 Test 2: parallel-plate capacitance

Model two superconducting plates separated by a dielectric. With $V_1 = V$ and $V_2 = 0$, the capacitance should approach

$$C = \frac{\varepsilon A}{d} \quad (25.1)$$

when fringing is negligible.

25.3 Test 3: floating island charge neutrality

Place a floating superconducting island near a fixed trapped-charge distribution. If the island has zero prescribed net charge, the solved island potential should adjust so that

$$Q_j = - \int_{\Gamma_j} \mathbf{n} \cdot \varepsilon \nabla \phi \, dA = 0. \quad (25.2)$$

This test verifies the global constraint implementation.

25.4 Test 4: screened-Poisson exponential decay

In a one-dimensional superconducting slab, impose a surface potential perturbation. The numerical solution should converge to

$$\phi(x) = \phi_0 e^{-x/\lambda_{\text{scr}}}. \quad (25.3)$$

This test should be used only when the mesh resolves λ_{scr} .

25.5 Test 5: offset-charge weighting potential

Compute w_j for an island and verify that direct trapped-charge solves and the weighting-potential integral give the same $Q_{j,\text{off}}$.

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